Digital Gift Card & Loyalty Point Exchange

Requirements Engineering & UML Use-Case Modelling — Solution

1. Requirements Table

Exactly 5 functional requirements and 2 non-functional requirements, following the lab requirements.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall validate merchant loyalty points, convert them into universal exchange tokens, and generate single-use digital voucher codes.	High	Pass: valid points produce tokens and a voucher with a cryptographic checksum. Fail: invalid or duplicate data is rejected.	Core conversion and voucher function.
FR-002	Functional	The system shall allow an account holder to view current loyalty-point and universal exchange-credit balances.	Medium	Pass: displayed balances match the ledger after a successful transaction. Fail: stale or inconsistent balance is shown.	Provides visibility of available value.
FR-003	Functional	The system shall apply the current dynamic exchange rate when converting eligible loyalty points into universal exchange credits.	High	Pass: the rate effective at transaction time is used and recorded. Fail: conversion is blocked if no valid rate exists.	Dynamic rates are a core platform function.
FR-004	Functional	The system shall allow an account holder to redeem a valid digital gift-card voucher using available universal exchange credits.	High	Pass: sufficient balance is debited once and voucher becomes redeemed. Fail: insufficient balance or invalid voucher prevents redemption.	Primary end-user value.
FR-005	Functional	The system shall lock a voucher before redemption and reject vouchers that are expired, already locked, or already redeemed.	High	Pass: only one redemption attempt obtains the lock. Fail: conflicting or invalid voucher is rejected.	Prevents duplicate redemption and fraud.
NFR-001	Nonfunctional	Point conversion and gift-card redemption operations must process within 250 ms with zero ledger balance drift.	High	Pass: tests meet ≤250 ms and reconciliation shows zero balance drift under simulated peak load.	Ensures performance and financial consistency.
NFR-002	Nonfunctional	The system shall protect voucher codes and transaction records using authenticated access and encrypted data in transit and at rest.	High	Pass: unauthorized access is denied and security tests confirm encryption.	Protects vouchers and transaction data.

3. Use-Case Flow Specification — UC-04 Redeem Gift Card

Use Case	UC-04 — Redeem Gift Card
Primary Actor	Account Holder
Supporting Actor	Merchant Partner
Preconditions	Account is authenticated; voucher is valid; sufficient universal exchange credits are available.
Postconditions	Credits are debited exactly once; voucher is marked redeemed; transaction is recorded.

Main Success Scenario

1. Account Holder selects **Redeem Gift Card** and enters/scans the voucher code.
2. System validates voucher status, expiry, and format.
3. System **locks the voucher** to prevent concurrent or duplicate redemption.
4. System checks the available universal exchange-credit balance.
5. System debits the required credits from the account ledger.
6. System marks the voucher as redeemed and records the transaction.
7. System displays confirmation and the updated balance.
8. Use case ends successfully.

Alternate Flow — Invalid / Already Used Voucher

3a. If the voucher is expired, already locked, or already redeemed, the system rejects the attempt.

3b. No exchange credits are debited.

3c. The system displays an error message and the use case ends without changing the account balance.